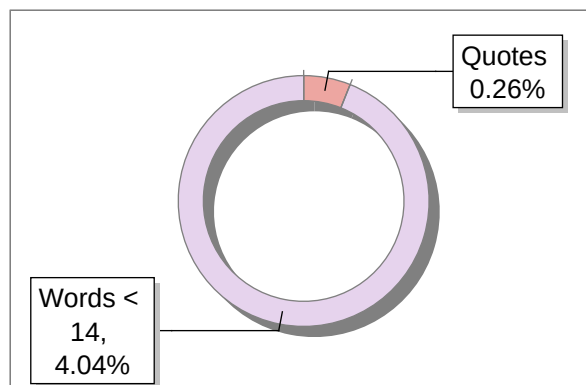
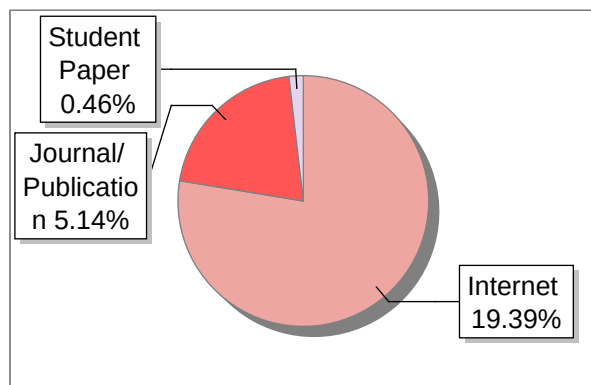


Submission Information

Author Name	Tanmay s ail and Tushan
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6	qdoc.tips	1	Internet Data
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29	www.mdpi.com	<1	Internet Data
30	Contributions to Economic Analysis Dynamic General Equilibrium Mode, by Dixon, Peter B. Ri- 2001	<1	Publication
31	ejournal.undip.ac.id	<1	Internet Data
32	hnlu.ac.in	<1	Publication

33	media.unpad.ac.id	<1	Publication
34	openlibrarypublications.telkomuniversity.ac.id	<1	Publication
35	www.topcounselingschools.org	<1	Internet Data

1. SYNOPSIS

1.1 Title of the project:

SKILL BITES

1.2 Introduction of project:

In today's modern era where technology has evolved and people started to rely more on online courses rather than going to colleges, we have come up with a user-friendly platform that makes online learning easy and engaging. This project focuses on using the latest tech to make personalized learning experiences. We work to make online education simple and accessible for everyone. This project aims to create an engaging and informative online course that helps learners achieve the desired learning outcomes.

1.3 Innovative idea behind the project:

This project aims to create a space where learners and instructors can connect seamlessly. It is created for every instructors who wants to provide knowledge to the students via recordings, courses and assignments and the instructor too get paid.

1.4 Nature of the project:

- Users can access a wide range of courses covering various subjects and skill levels.
- Users can enhance their skills in specific areas through targeted courses and exercises.
- Simplify tasks for instructors and promote efficiency in grading, announcements and enrolment.
- Users can track their progress within courses, set goals, and monitor their achievements over time.
- Incorporate interactive elements, such as assignments, and discussions.
- Save users time and money by eliminating the need for travel and providing cost-effective learning alternatives.

1.5 Literature Survey:

Conducted survey on existing course-based website on the internet to understand the features, functionality and design. Api's related to real time communication, payment gateways.

1.6 Hardware Software Requirements:

- **Software requirements:**

Platform: Microsoft windows

IDE: Visual Studio code

Frontend Technology: html, css, javascript, ReactJs

Backend Technology: MongoDB database, Node.js

Additional libraries and dependencies as per the project requirements.

- **Hardware requirements:**

Adequate storage for storing course data (MongoDB)

Cloudinary to store videos and images

This project holds immense potential to become a go-to destination for e learning seeking an ultimate knowledge and education.

1.7 Team Members

Tanmay s ail

Tushan

2. SOFTWARE REQUIREMENT SPECIFICATION

2.1 Introduction:

The Software Requirement Analysis document provides a complete overview description of all the functionalities and specification of the “SkillBites” system. The SRS is a comprehensive document that outlines the functional and non-functional requirements of a software system.

2.2 Purpose:

This document is developed for the project “SkillBites” to provide a platform for the instructors who is willing to start a paid online course to the students who want to learn specific courses. Here the students can interact with the instructors and can go through the posted assignments, study materials, quiz and recorded class. An SRS minimizes the time and effort required by the developers to achieve desired goals and also minimizes the development cost.

2.3 Scope:

The main objective of the project “SkillBites” which is designed to provide a platform to students to buy courses online and study online.

It provides the following features

- The instructors have the capability to create courses with reasonable price, providing a diverse range of learning opportunities. Admin can view Instructor created course
- Students can post queries regarding any assignments.
- Instructor can assign assignment for their students The system captures and maintains records of various aspects of student performance. This includes details about assignment submissions, quiz results, and overall performance. This feature enables both students and instructors to monitor and evaluate progress.
- Students can enroll the courses they want
- Students have the option to post queries related to assignments.
- Upon successful completion of a course, students will receive a certificate of completion.

2.4 Defination, Acronnys, abbreviations

ADMIN - THE ADMINISTRATOR

CPU -CENTRAL PROCESSING UNIT

JS- JAVASCRIPT

GUI - GRAPHICAL USER INTERFACE

HTML -HYPER TEXT MARKUP LANGUAGE

CSS -CASCADING STYLE SHEET

DBMS -DATABASE MANAGEMENT SYSTEM

UI -USER INTERFACE

SRS - SOFTWARE REQUIREMENT SPECIFICATION

2.5 Overview:

The project "Skill Bites" is a website-based initiative aimed at facilitating instructors in offering paid online courses to students interested in learning specific subjects. The Software Requirements Specification (SRS) document serves as a comprehensive guide outlining all the necessary requirements for the successful development and implementation of this project.

2.6 Overall Description:

This section will give an overview of the whole system. It will also describe the various users who will use the system and what type of functionality is available for those users. The general, functional, performance and security requirements will be presented.

2.7 Product Perspective

The "Skill Bites" project envisions the creation and maintenance of an inclusive online learning platform. The users of the system are categorized into three categories which are admin, instructor and student. Every user should be given an online connection. The platform is designed with a user-friendly interface to ensure ease of navigation for both instructors and students, promoting a seamless learning experience.

2.8 Product Functioning:

- The project "Skill Bites" is developed to provide an online platform for admin, student and Instructors where the admin can manage the other two users, that is the student and instructor.
- The product first asks if the user wants to join in as a student or an instructor.
- Internet connection is mandatory.
- The system enables the instructor to create course and waits for the approval of the admin to add it to the course lists.
- It enables the instructors to post study materials, assignments and quizzes.
- Students can access the study materials and other assessments posted by the instructors of particular course.
- Students can post their queries on recorded videos or to clear any doubts.
- Instructors can get 90% of income while the remaining goes to the admin.

2.9 Browser That Supports:

- Google chrome
- Microsoft edge
- Internet explorer
- Mozilla
- Firefox

2.10 User classes and characteristics:

The system has three levels of users.

Admin:

- Login to admin platform
- Admin can manage the students
- It manages the instructor
- Manage the courses
- Manage feedbacks on the website
- Manage reports on courses students and instructor

Instructor:

- Instructor registration and login
- They can also manage their profile
- Create courses by entering details of the specific course
- Upload assignments
- They can manage attendance of each student
- They can upload video recordings
- Upload study materials
- They can reply to the queries posted by the students
- They can view the student's assignments

Student:

- Student registration and login
- They can manage their profile
- View all courses
- Purchase specified courses
- View instructor created recordings by either downloading or watching online
- They can ask queries to instructor

2.11 General Constraints:

- The web application requires an internet connection
- Users must register themselves
- Nobody has the right to change information of someone else account

2.12 Assumptions and Dependencies:

- It is ³¹assumed that there are three levels of users, i.e., admin, student and instructor.
- All the data entered will be correct and up to date.
- The system assumes the implementation of security measures to protect user data, such as login credentials and personal information, from unauthorized access or breaches.

2.13 Specific Requirements:

User Interface:

- Textboxes to enter information.
- Buttons to create, add, delete, update and search
- Labels to display the information
- Checkboxes and radio buttons
- Dropdowns

Software Requirement:

- Operating system - Windows
- Text Editor - Visual studio code
- Language - NodeJS
- UI - HTML, CSS, JS, ReactJS
- Database -MongoDB
- Server -Nodejs server
- Browser -Chrome, Microsoft edge,Mozilla firefox

Hardware Requirements:

- RAM -4gb
- Processor -Intel core i3 or above
- Hard disk -minimum 64gb

2.14 Communication Interface:

This is a web-based system and communication is done through the internet and internet protocols and http protocols.

2.15 Functional Requirements:

Admin:

Login module: Admin can enter the website using his unique email-id and password

Dashboard: Admin can view the number of students and instructors registered in the system.

Manage student: Admin manages the students who have registered to the system.

Manage Faculties: Admin manages the instructors who have registered to the system.

Manage Feedbacks: Admin can manage feedbacks by student or instructors.

Manage Payments: Admin can manage payment related activities.

Student:

Student Registration and login: Students can enroll themselves to the system as a student with username, email-address, password.

Manage Profile: Students can manage their profile within the system. This includes the ability to update and modify their personal information.

View Courses: Students can access a list of available courses in the system.

View bought course: Students can view a list of their acquired courses for reference and tracking.

Assignments: Students can likely access and complete assignments related to their enrolled courses.

Lectures: Students can access a list of recordings provided by the instructor.

Queries: Students can ask questions or seek clarification on course-related matters.

Feedbacks: Students can provide feedback on the courses or the system itself.

Instructors:

Instructor registration and login: Instructors have the ability to register and log in to the system as an instructor.

Manage Course: Instructors can manage details related to the courses they are responsible for tasks such as creating new courses, updating course content, viewing student performance and much more.

Add Assignments: Instructors have the capability to create and add assignments to the courses they are teaching.

Add recordings: They can also add video recordings and post links for more information.

Provide marks: Instructors can provide marks for students who submits the assignments.

Reply to Feedbacks: Instructors can respond to feedback provided by students, addressing comments, concerns, or suggestions submitted by students in the form of feedback.

2.16 Performance Requirement:

The server should be able to support an indefinite number of active students and courses without compromising performance or losing data. Ensures that users, including students and instructors, can quickly access and navigate the platform without extended delays.

2.17 Design constraints:

- All the inputs should be checked and validated.
- Details given by the user during the registration should be stored in the database.
- Details entered by the admin and user should be stored in database.
- Alert the users for incorrect data given to an input field.

2.18 System Attributes:

- **Performance:** The system executes tasks and processes data, ensuring optimal response times and resource utilization.
- **Reliability:** A reliable system ensures that users can depend on its functionality, minimizing errors, crashes, and disruptions.
- **User-Friendly Interfaces:** The design and usability of the system must be intuitive, easy to navigate, and user-friendly.
- **Maintainability:** The system can be maintained, updated, and modified over time.
- **Portability:** The system must be easily transferred or adapted to different environments or platforms.
- **Flexibility:** The system must accommodate changes, updates, or modifications without significant disruption.
- **Timeliness:** The system must meet the specified deadlines and response time requirements.

2.19 Safety and Security Requirement:

Admin has all rights to the system. Email id should be unique around the system and cannot be used twice for different users. Only the authenticated users can access the system. The server on which the project resides will have security to prevent unauthorized access.

3.SYSTEM DESIGN

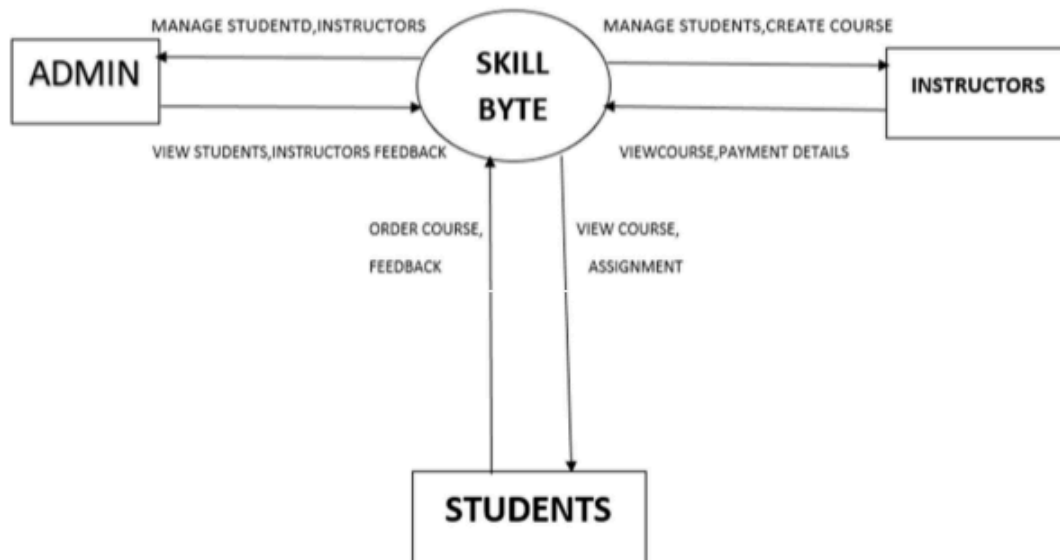
3.1 Introduction

The goal of the design process is to produce a model or representation of a system, which can be used later to build the system. System design is the process of defining the architecture, modules, interfaces, and data for a system to satisfy specified requirements. The focus of system design is on deciding which modules are needed for the system, the specifications of these modules, and how the modules should be interconnected.

This crucial process translates user needs into a blueprint, a model that guides the system's construction. Systems design involves defining the system's architecture, its building blocks (modules), the communication channels (interfaces) between them, and the data it handles. The core focus is on identifying the necessary modules, their detailed functionalities, and how they seamlessly interact to achieve the desired outcome.

3.2 Context Flow Diagram

A context flow diagram is a top-level data flow diagram. It only contains one process node that generalizes the function of the entire system in relationship to external entities. In the context diagram, the entire system is treated as a single process and all its inputs, outputs, sinks, and sources are identified and shown.



3.3 Data flow diagram:

Data Flow Diagrams (DFDs) are like X-rays for systems, revealing the inner workings of data movement. These visual roadmaps depict how information flows through a system, from external sources and destinations (think entry and exit points) to internal processing hubs. Using a set of standardized symbols, DFDs map the transformations of data as it progresses through various stages. This graphical technique, also known as bubble charts or data flow graphs, offers a powerful tool for understanding systems at any level of complexity. By decomposing a system into progressively detailed layers, DFDs provide a clear view of information flow and pinpoint areas for optimization.

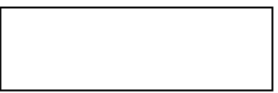
DFD is also known as bubble charts or Data Flow Graphs. DFD may be used to represent the system at any level of abstraction. DFDs may partition into a level that represents increasing information flows and functional details.

3.4 Rules Regarding DFD Construction:

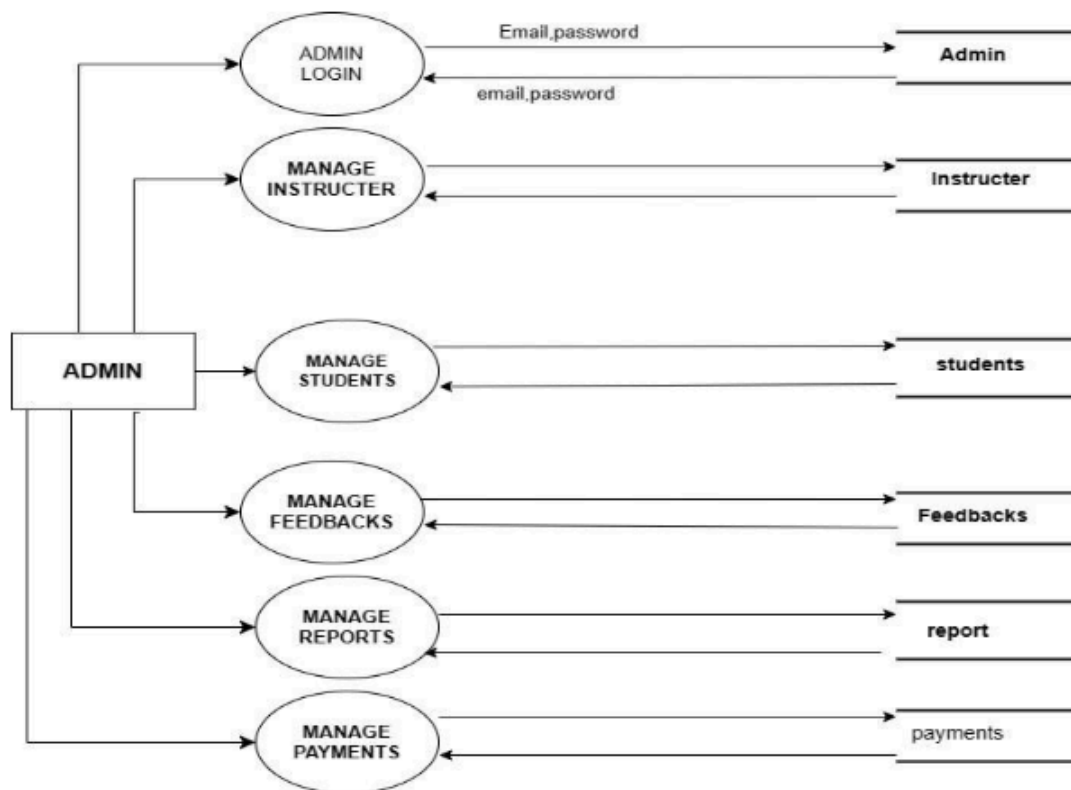
- A process cannot have only outputs.
- A process cannot have only inputs.
- The inputs to a process must be sufficient to produce the outputs from the process.
- All data stores must be connected to at least one process.
- All data stores must be connected to a source or sink.
- A data flow can have only one direction of flow. Multiple data flows to and/or from the same process and the datastore must be shown by separate arrows.
- If the same data flows to two separate arrows, it should be represented by a forked arrow.
- Data cannot flow directly back into the process it has just left. All data flows must be named using a noun phrase.

3.5 DFD Symbols:

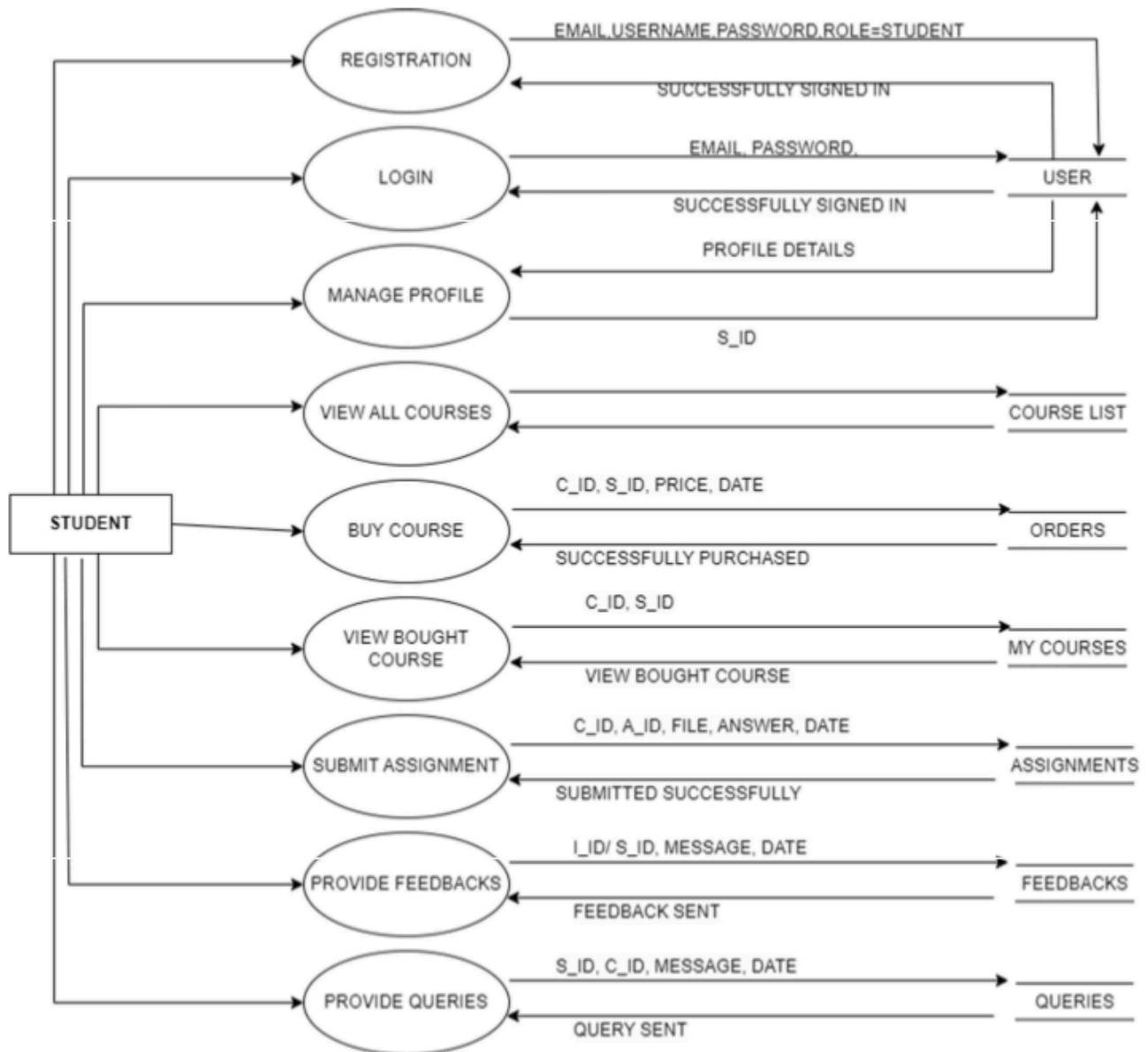
Name	Notation	Description
Process		A process transforms incoming data flow into outgoing data flow. The processes are shown by named circles.
Datastore		Datastores are repositories of data in the system. They are sometimes also referred to as files.

Dataflows		5 Data flows are pipelines through which packets of information flow. Label the arrows with the name of the data that moves through them.
External entity		External entities are objects outside the system with which the system communicates. External Entities are sources and destinations of the system's inputs and outputs

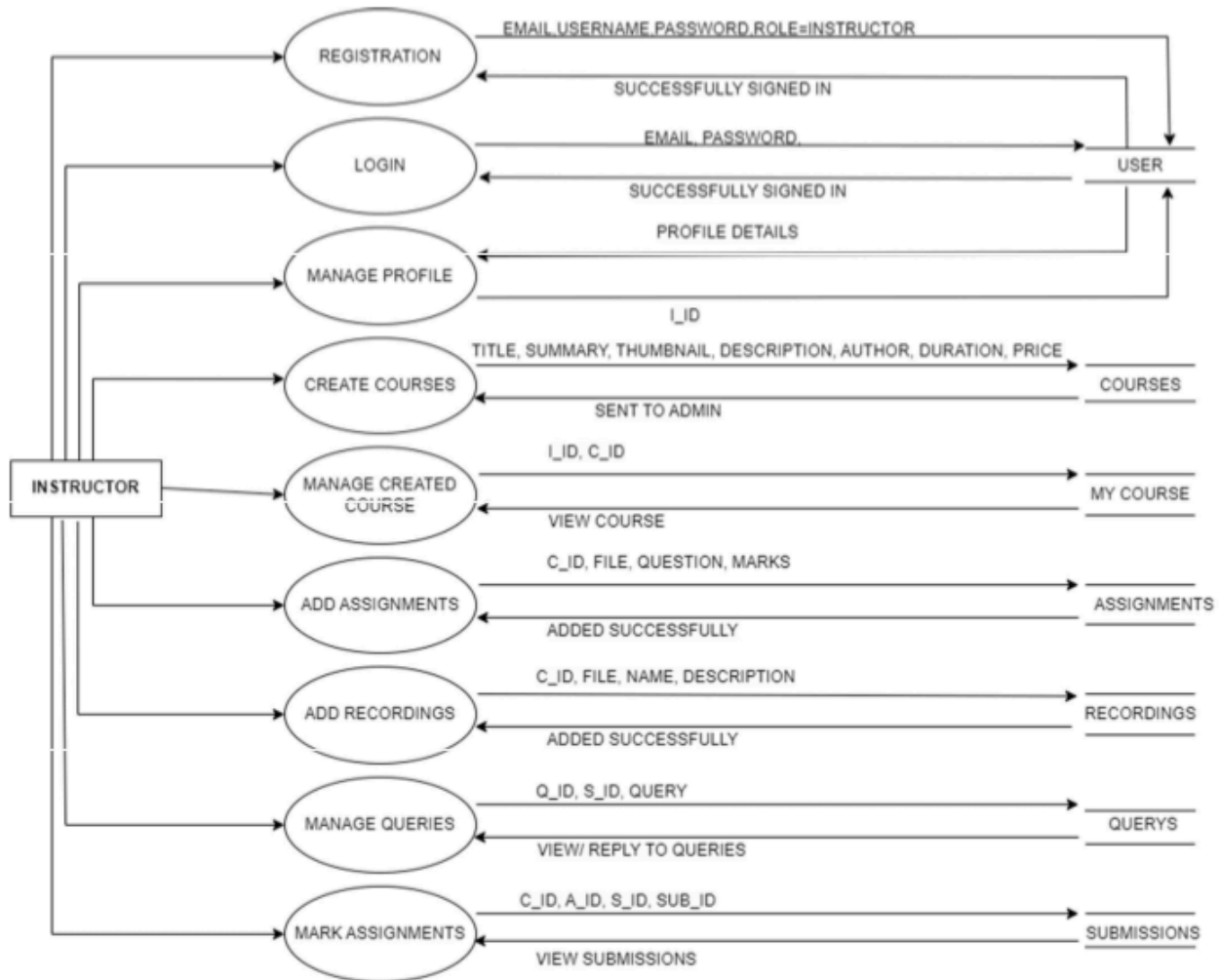
3.6 DFD LEVEL 1 : ADMIN



3.7 DFD LEVEL 1 : STUDENT



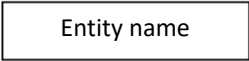
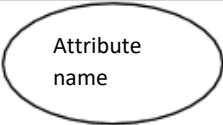
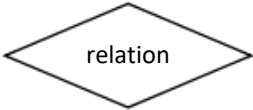
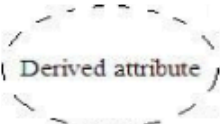
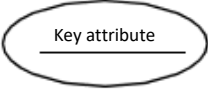
3.8 DFD LEVEL 1 : INSTRUCTOR



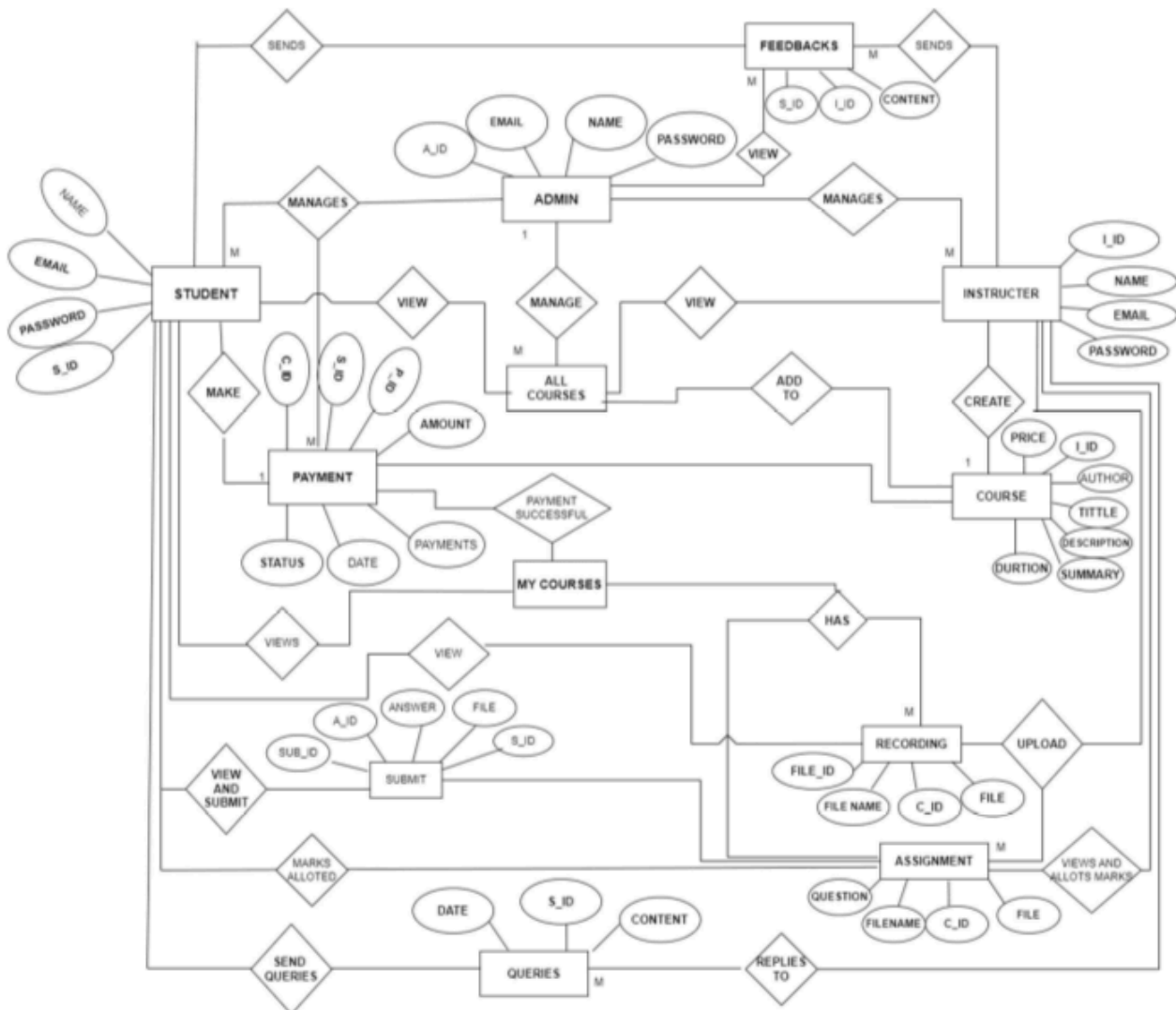
3.9 Entity-Relationship Diagram:

The basic objective of the ER model representation is an entity that is a “thing” in the real world with an independent existence. Entities are physical items or aggregations of data items that are important to the business we analyze or to the system; we intend to build. An entity represents an object defined within the information system about which you want to store information. Entities are named as singular nouns and are shown in rectangles in an ER-Diagram. Each entity is described by several attributes; individual instances of an entity will have different attribute values.

3.10 ER-Diagram Symbols:

Name	Notation	Description
Entity		It may be an object with a physical existence or conceptual existence. It is represented by a Rectangle.
Attribute		The properties of the entity can be an attribute. It is represented by an Ellipse.
Relationship		Whenever an attribute of one entity refers to another entity, some relationship exists. It is represented by a Diamond.
Link		Lines link attributes to entity sets and entity sets to relations.
Derived Attribute		A dashed ellipse denotes derived attributes
Key Attribute		An entity type usually has an attribute whose values are distinct for each entry in the entity set. It is represented by an Underlined word in an ellipse.
Multivalued Attribute		Attributes that have different numbers of values for a particular attribute. It is represented by a Double ellipse that represents multi-valued attributes.
Cardinality Ratio	1. 1:1 2. 1:M 3. M:1 4. M:M	It specifies the maximum number of relationship instances that an entity can participate in. There are four cardinality ratios.

3.11 ER DIAGRAM:



4. DATABASE DESIGN

4.1 Introduction:

The database design is the most crucial part of the SkillBites system. It involves defining the structure of the database, the types of data to be stored. It also establishes relationship between a set of data's. An effective database design ensures storage of data, retrieval and manipulation of data. The SkillBites system use a nosql database which is the MongoDB database where the data is stored in documents.

4.2 The data for “Skill Bites” is organized into 4 documents with sub documents

- Users
- Courses
 - Enrolled
 - Recordings
 - Assignments
 - submit
- Purchases
- Feedbacks

4.3 Database Structure

4.3.1 Structure of Users document

Field name	Type	Constraints	Description
u_id	ObjectId	required	Automatically assigns an object Id
username	String	required	Username of the user
email	String	unique	Email should be unique
password	String	required	Password
role	String	Enum['Admin','Instructor','Student']	Role of users
bio	String		Biography of user
phone	Number		Mobile number of the user
address	String		Address of the user
photo	String		Profile photo
timestamps	Date		Created and updated date

4.3.2 Structure of Course document

Field name	Type	Constraints	Description
c_id	ObjectId		Automatically assigns an objectId
title	String	required	Title of the course
summary	String	required	Brief summary
content	String	required	Description
cover	String	required	Thumbnail
duration	Number	required	Duration in months
price	Number	required	Total Price of the course
author	ObjectId(Users)	required	Refers the instructor who created the course
total	Number	Default:0	Total marks of course including the marks of assignments and recordings
enrolled	Array of embedded objects		List of enrolled students
approved	Boolean	Default:false	Status of approval of course from the admin
recordings	Array of embedded objects		List of recordings
assignments	Array of embedded objects		List of assignments
timestamps	Date		Created and updated date

4.3.2.1 Embedded Object: enrolled

Field name	Type	Constraints	Description
enroll_id	ObjectId		Automatically assigns an object Id
student	ObjectId (Users)	required	Refers the student id of the enrolled student
totalMarks	Number	Default:0	Total marks of student in a course

4.3.2.2 Embedded Object: recordings

Field name	Type	Constraints	Description
Rec_id	ObjectId		Automatically assigns an object Id
file	String	required	Lecture file
filename	String	required	Title of the lecture
description	String	required	Lecture details and also provide links
watched	Array of ObjectId(Users)		Contains student id of students who watched the lecture
createdAt	Date	Default:Date.now()	Created date

4.3.2.3 Embedded Object: Assignments

Field name	Type	Constraints	Description
Assign_id	ObjectId		Automatically assigns an object Id
title	String	required	Title of the assignment
Question	String	required	Question of the assignment
file	String	-	File attached to the assignment
marks	Number		Marks of the assignment
createdAt	Date	Default:Date.now()	Created date
submit	Array of embedded Objects		List of submissions

4.3.2.3.1 Embedded Object: submit

Field name	Type	Constraints	Description
Submit_id	ObjectId		Automatically assigns an object Id
student	ObjectId(Users)	required	Refer the student id of the student who submitted the assignment
file	String	-	File attached to the submission
description	String	-	Answer of the student
marks	Number		Marks allotted to the student if status is submitted
status	String	Enum['submitted', 'pending']	Created date
submittedAt	Date	Default:Now	Submitted date

4.3.3 Structure of Purchases document

Field name	Type	Constraints	Description
Purchase_id	ObjectId		Automatically assigns an object Id
userId	ObjectId(Users)	required	Refer the student id of the student who bought the course
courseId	ObjectId(Courses)	required	Refer the course id the particular student bought
paymentId	String	required	Payment id

4.3.4 Structure of Feedbacks document

Field name	Type	Constraints	Description
f_id	ObjectId		Automatically assigns an object Id
name	ObjectId(User)	required	Refer the id of the user who sends the feedback
email	String	required	Email of the user
message	String	required	Message contents
timestamps			Sent date

5. Detailed Design

5.1 Introduction:

The purpose of preparing this document is to explain the complete design in detail of the Online classroom management system. This detailed design report will mainly contain the general definition and features of the project, design constraints, the overall system architecture, and data architecture. Additionally, a brief explanation of our current progress and schedule of the project will be provided in related sections. The design of the system and subsystem modules will be explained both verbally and visually using diagrams to help the programmer to understand all information stated in this document correctly and easily.

5.2 Application documents:

The documents used during the detailed design are :

- System Requirements Document
- System Design
- Database Design

5.3 Structure of the software package

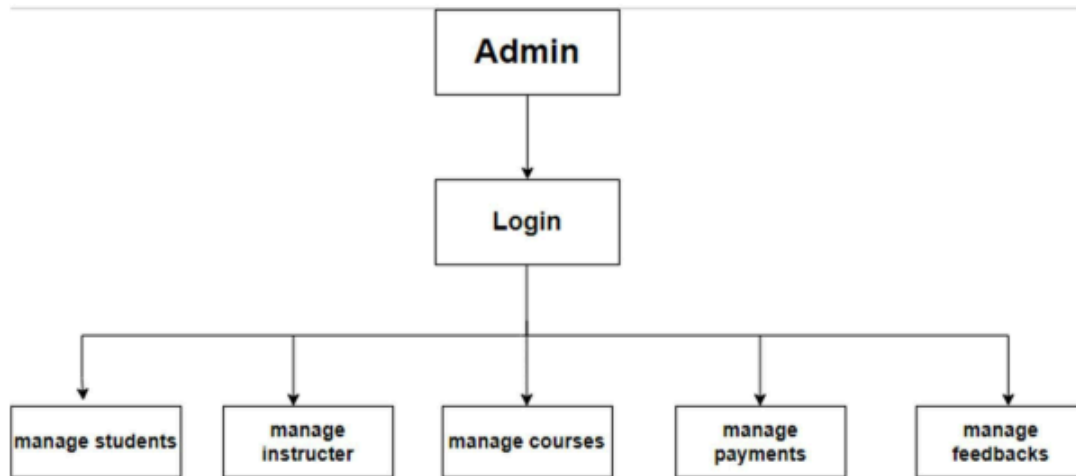
The components are:

- Registration component for Student and Instructor.
- Login component for Admin, Instructor and Student.
- Course component for Instructor, Student and Admin.
- Purchase component for Student.
- Assignment and recording components for Instructor and students.

5.4 Modular Decomposition of Components

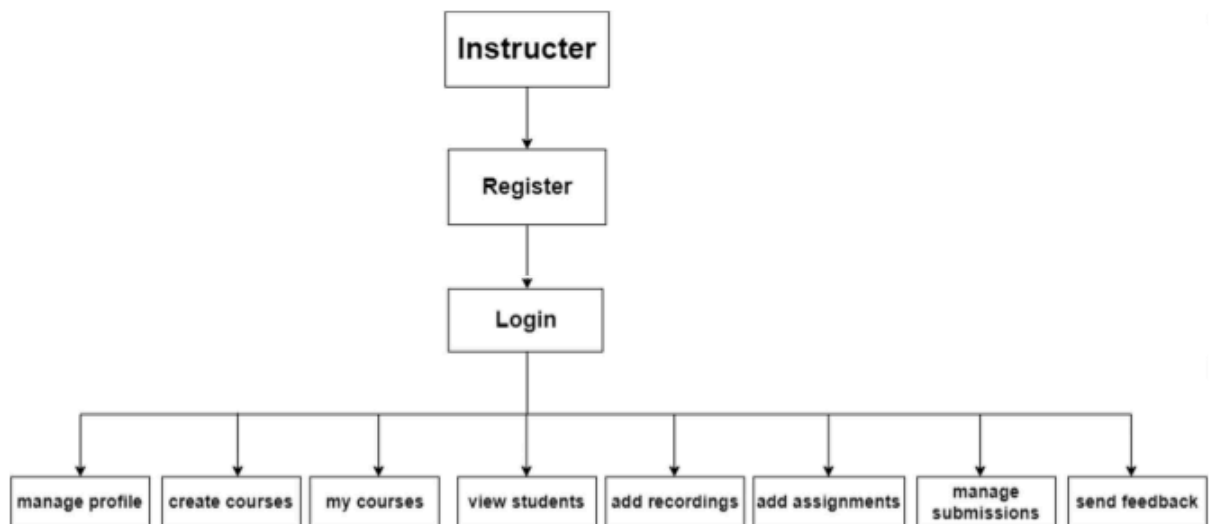
5.4.1 Admin Component:

5.4.1.1 Structure chart for Admin.



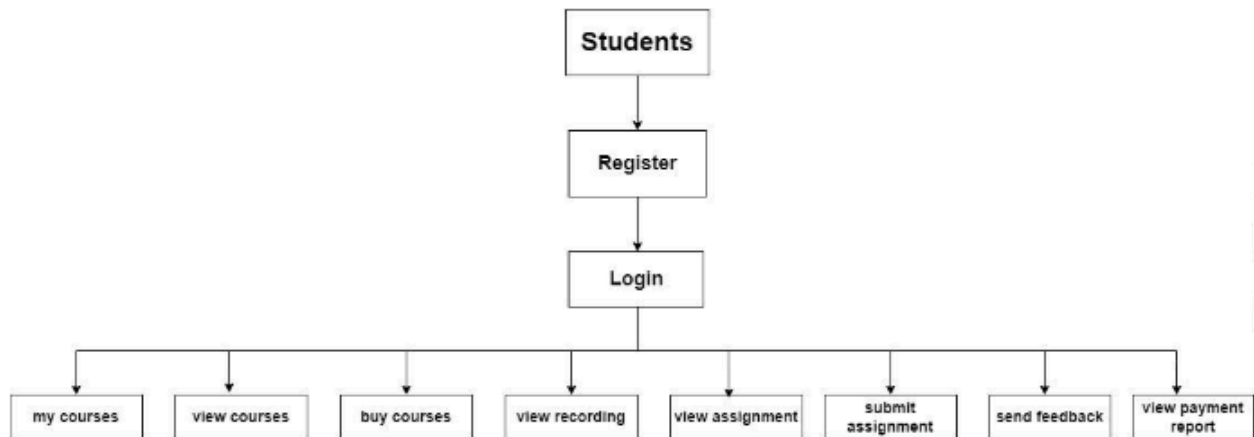
5.4.2 Instructor Component:

5.4.2.1 Structure chart for Instructor.



5.4.3 Student Component:

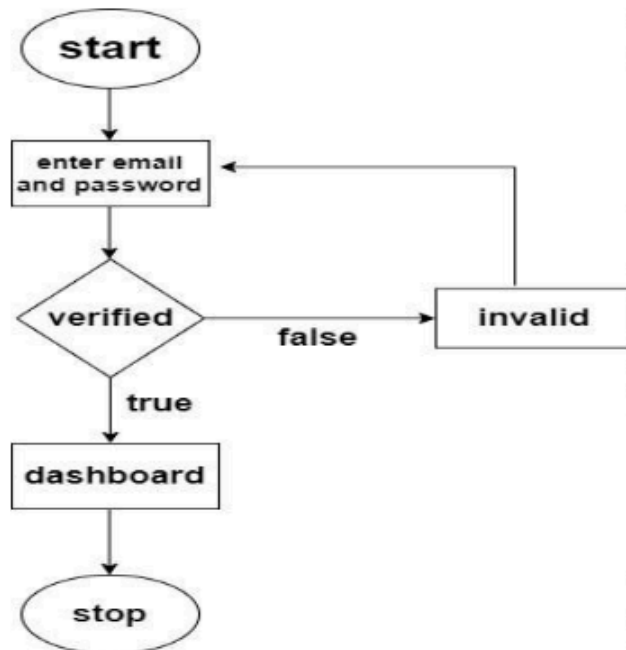
5.4.3.1 Structure chart for Student.



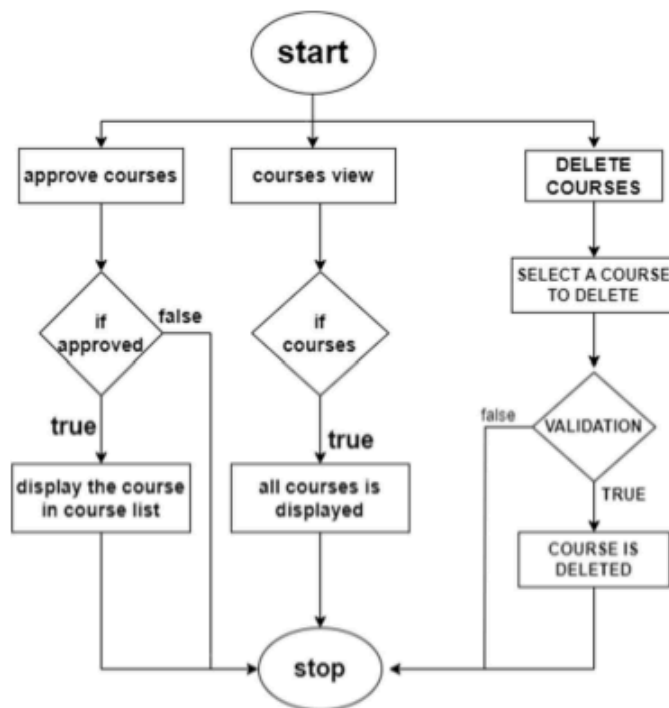
5.4.4 Procedural details for Admin

Login:

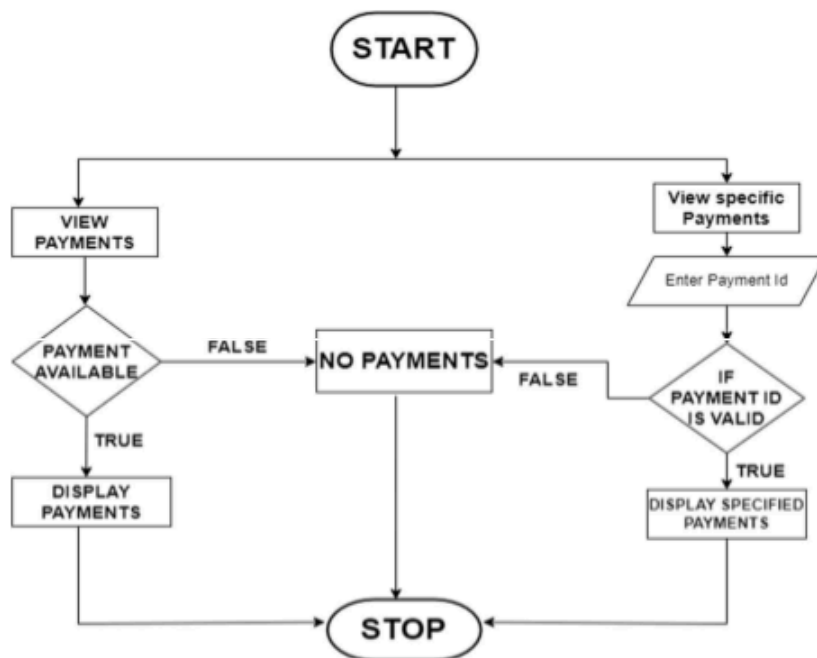
Input: Email, password



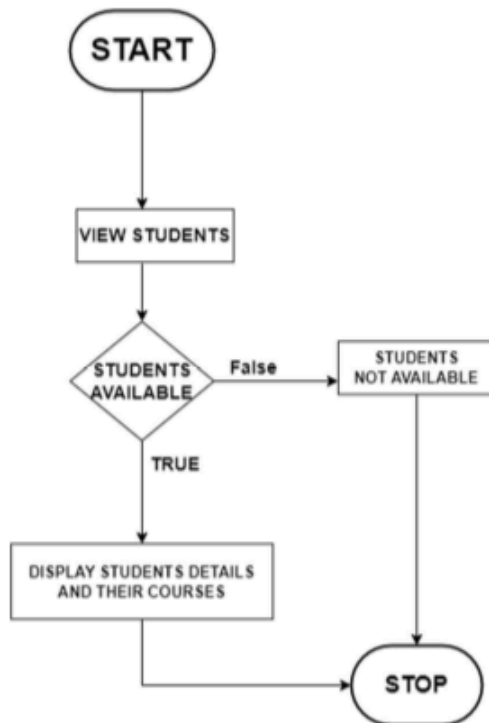
Manage courses:



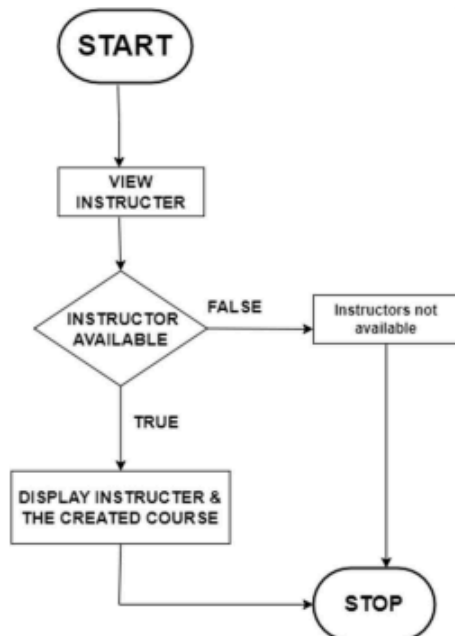
Manage Payments:



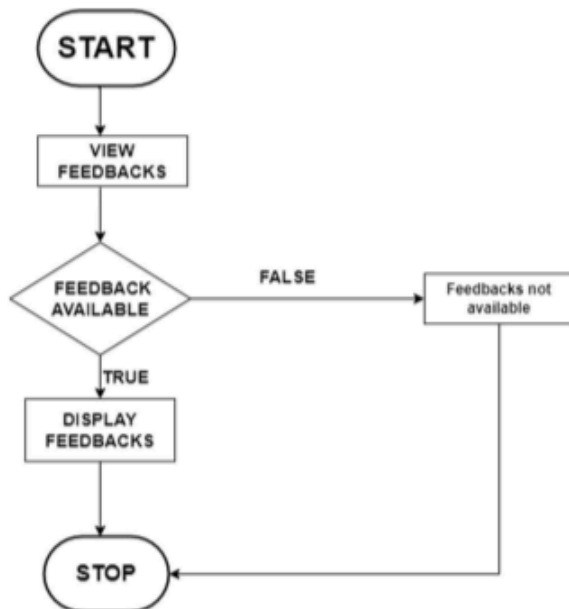
Manage Students:



Manage Instructors:

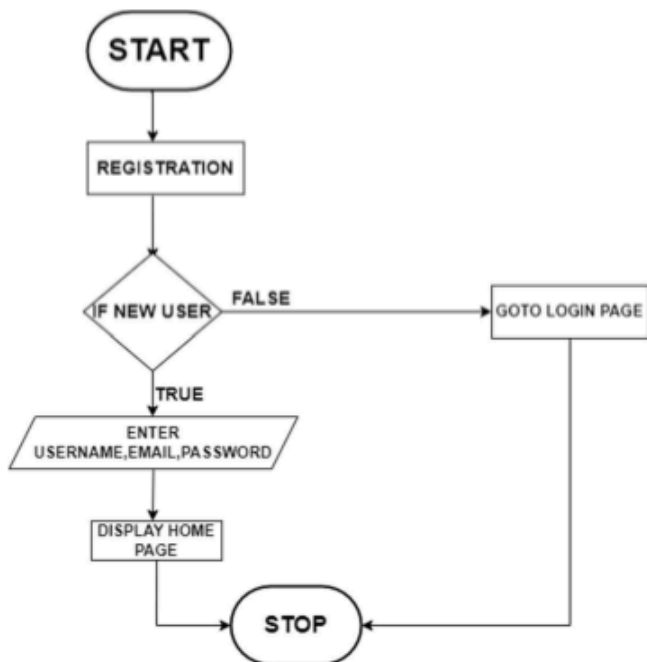


View Feedbacks:

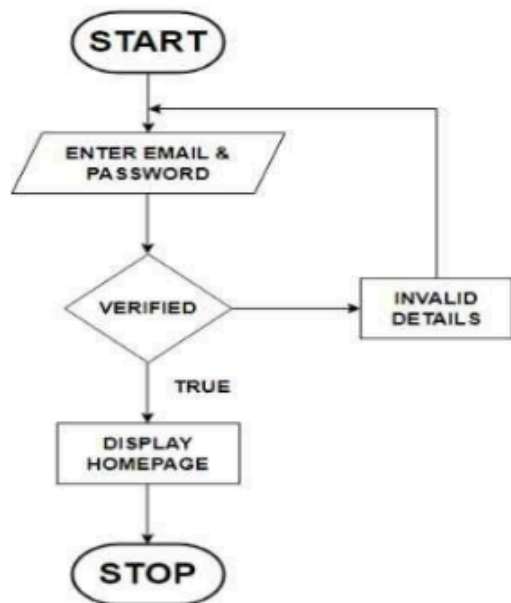


5.4.5 PROCEDURAL DETAILS OF INSTRUCTORS:

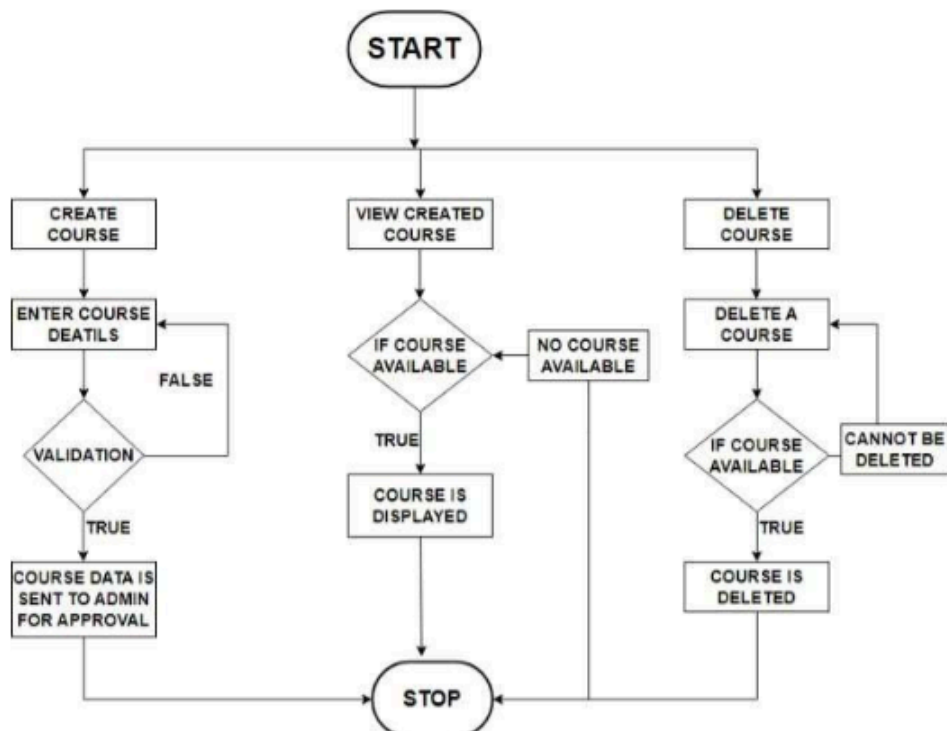
Registration:



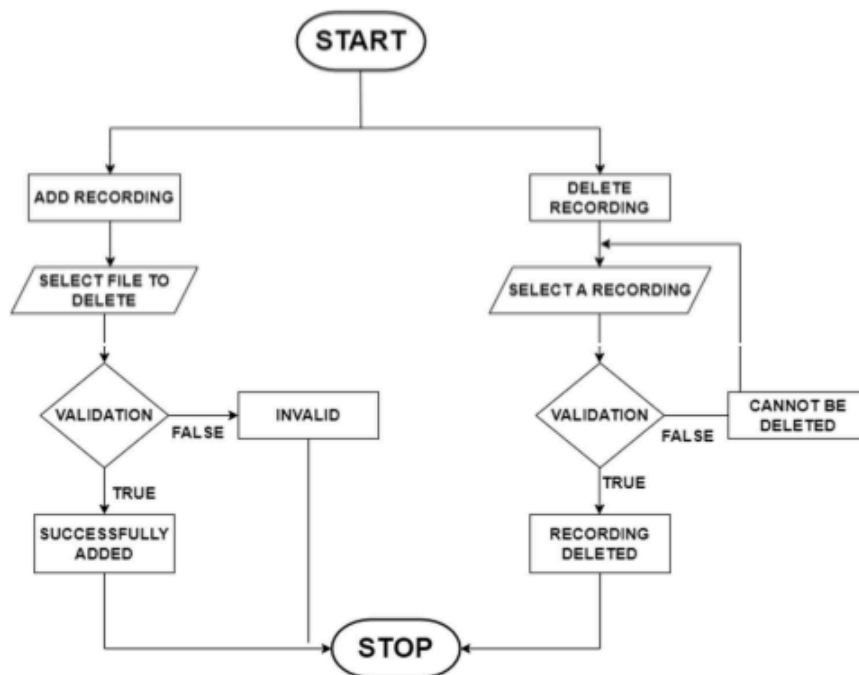
Login:



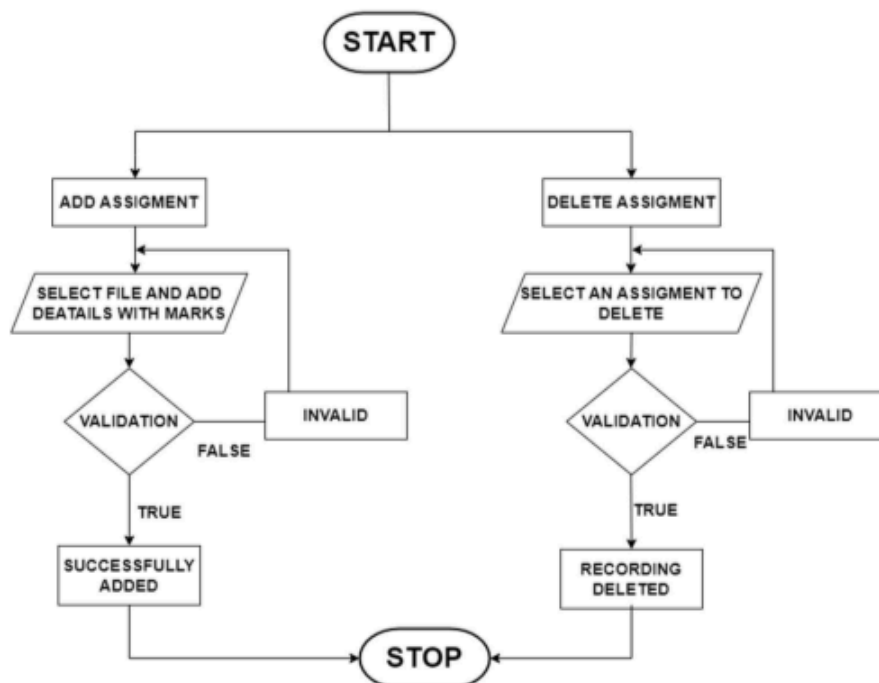
Manage Course:



Manage Course Recordings:



Manage Assignments:

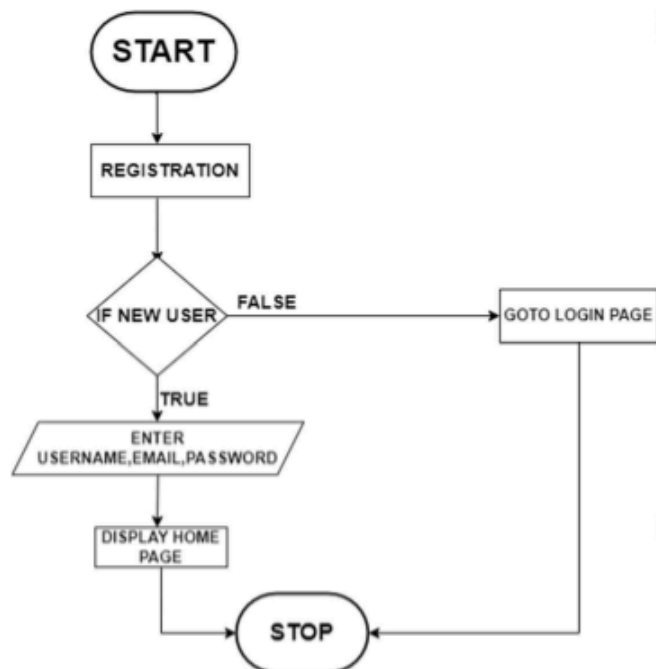


Manage Submissions:

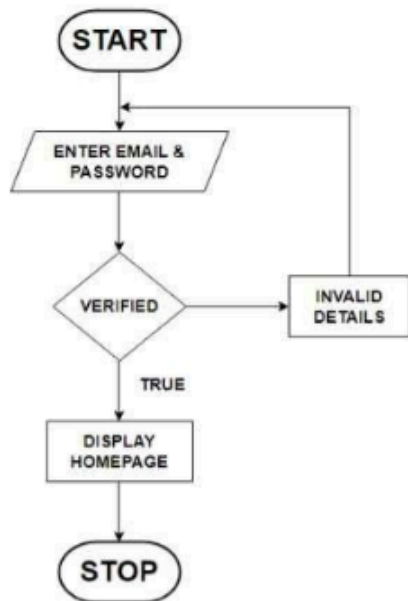


5.4.6 PROCEDURAL DETAILS OF STUDENTS:

REGISTRATION:



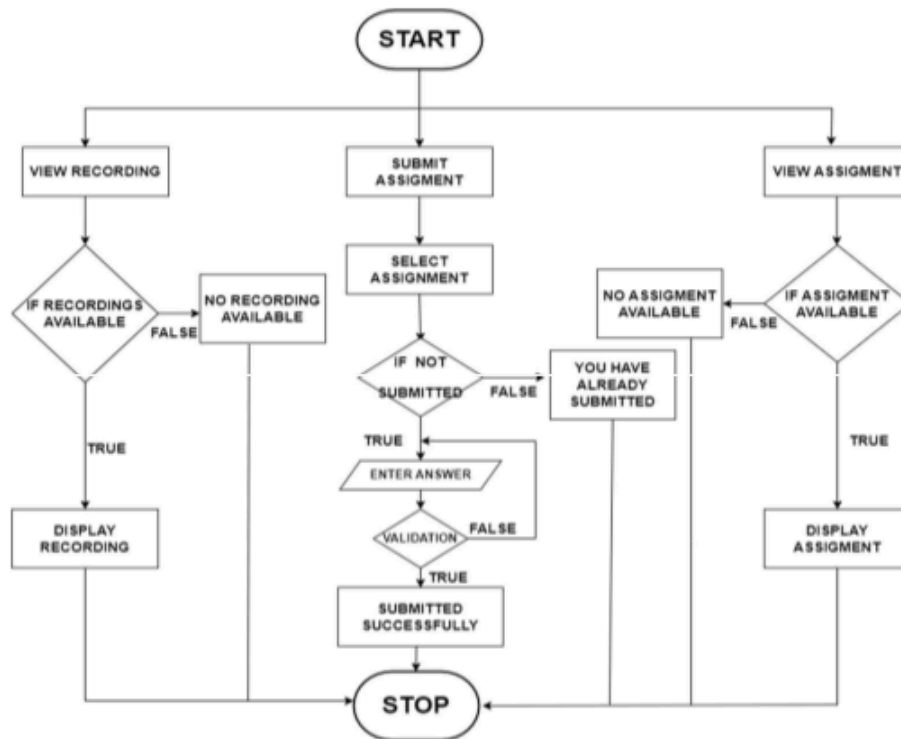
LOGIN:



Make Payments and view Purchased Course:



View Recordings, Assignments and Submit Assignment



Get Certificate



7. Testing

7.1 Introduction

In this ²⁵ software testing has been defined as ¹⁸ the process of analysing a software item to detect the differences between existing and required conditions and to evaluate the features of the software item. ¹² Software testing is the process used to assess the quality of computer software. It involves operation of a system or application under controlled conditions and evaluating the results. The controlled conditions should include both normal and abnormal conditions. Testing should intentionally attempt to make things go wrong to determine if things happen when they should.

7.2 Testing Objectives

- ⁹ Finding defects which may get created by the programmer while developing the software.
- Gaining confidence in and providing information about the level of quality
- To prevent defects
- ¹¹ To make sure that the end results meets the business and user requirements.
- To ensure that it satisfies the BRS that is Business Requirement Specification and SRS that is System Requirement Specification.

7.3 Testing steps

7.3.1 ¹ Unit Testing

Unit testing focuses efforts on the smallest unit of software design. This is known as module testing. The modules are tested separately. The test is carried out during programming stage itself. In this step, each module is found to be working satisfactory as regards to the expected output from the module.

7.3.2 Integration Testing

Data can be lost across an interface. One module can have an adverse effect on another, sub functions, when combined, may not be linked in desired manner in major functions.

Integration testing is a systematic approach for constructing the program structure, while at the same time conducting test to uncover errors associated within the interface. The objective is to take unit tested modules and builds program structure. All the modules are combined and tested as a whole.

7.3.3 ¹⁷ Validation Testing

¹ At the culmination of the integration testing, Software is completely assembled as a package. Interfacing errors have been uncovered and corrected and a final series of software test begin in validation testing. Validation testing can be defined in many ways, but a simple definition is that the validation succeeds when the software functions in a manner that is expected by

the customer. After validation test has been conducted, one of the three possible conditions exists.

- The function or performance characteristics confirm to specification and are accepted.
- A deviation from specification is uncovered and a deficiency lists is created.
- Proposed system under consideration has been tested by using validation test and found to be working satisfactory.

7.3.4 Output Testing

After performing the validation testing, the next step is output testing of the proposed system, since no system could be useful if it does not produce the required output in a specific format. The output format on the screen is found to be correct. The format was designed in the system design time according to the user needs. Also for hard copy the output comes as per the specified requirements by the user. Hence output testing did not result in any correction for the system.

7.3.5 User Acceptance Testing

User acceptance of a system is the key factor for the success of any system. The system under consideration is tested for the user acceptance by constantly keeping in touch with the prospective system users at the time of developing and making changes whenever required. This is done in regard to the following point:

- Input screen design.
- Output screen design.
- Online message should be guide to the user.

Login

Input	Test condition	Test output	Comments
Login	If the admin/ Instructor/Student email_id is not entered	Please fill out this field.	Username is required
	If the password is not entered	Please fill out this field	Password is required
	If the username and password are not valid	User not found	Enter valid username and password
	If the student/Instructor email_id and password are valid.	Successfully logged in.	Appropriate password and email is entered and the users are now logged in.
	If the admin email_id and password are valid.	Successfully logged in.	Appropriate password and email is entered and the Admin is now logged in to the

			admin page.
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Register

Input	Test condition	Test output	Comments
Register	If the username is not entered	Please fill out this field.	Username is required
	If the email_id is not entered	Please fill out this field.	Email is required
	If the entered email_id exists	User already exists	Register through another email_id
	If the password is not entered	Please fill out this field. Minimum 8 letters required	Password required
	If the above fields are valid.	Registered successfully	User registered

7.4 Integration Testing

Admin Login

Test No.	Test condition	Test output	Expected output
1	Admin login	The site is redirected to admin page	Success

Instructor/Student Register and login page

Test No.	Test condition	Test output	Expected output
1	Register	The entered data is stored in database and the site is redirected to home page	Success
2	Login	The site is redirected to homepage	Success

Admin Index page

Test No.	Test condition	Test output	Expected output
1	Dashboard	The site is redirected to Admin Dashboard page	Success
2	Manage Students	The site is redirected to Students list page	Success
3	Manage Instructors	The site is redirected to Instructors list page	Success

4	Manage Courses	The site is redirected to Courses list page	Success
5	View Feedbacks	The site is redirected to feedbacks list page	Success

Student Index page

Test No.	Test condition	Test output	Expected output
1	Manage Profile	The site is redirected to Student Profile page	Success
2	View Course List	The site is redirected to all Courses page	Success
3	View Courses	The site is redirected to Purchased courses page to view course content	Success
4	View purchases	The site is redirected to purchases page to view the order details information	Success
5	Logout	The site is redirected to homepage	Success

Instructor Index page

Test No.	Test condition	Test output	Expected output
1	Manage Profile	The site is redirected to Instructor Profile page	Success
2	View Course List	The site is redirected to all Courses page	Success
3	Create Course	The site is redirected to Create Course page	Success
4	View Courses	The site is redirected to view created course page	Success
5	Logout	The site is redirected to homepage	Success

View Course page (Instructor)

Test No.	Test condition	Test output	Expected output
1	Add assignment	Opens the add Assignment modal	Success

2	Add recordings	Opens the add Recordings modal	Success
3	View student assignments	The site is redirected to Student assignments page	Success
4	View lectures	The site is redirected to view lecture page	Success
5	View student performance	The site is redirected to performance page	Success

View Course Page (Student)

Test No.	Test condition	Test output	Expected output
1	View lectures	The site is redirected to view lectures page	Success
2	View Assignments	The site is redirected to view assignments page	Success
3	Submit assignment	Opens the submission box	Success
4	View course performance	The site is redirected to performance page	Success

7.5 System Testing Table

System Testing is the testing of a complete and fully integrated software product. Usually, software is only one element of a larger computer-based system. Ultimately, software is interfaced with other software/hardware systems. System Testing is actually a series of different tests whose sole purpose is to exercise the full computer-based system.

Sl.No	Test condition	Test report
1	System lading	Successful
2	System run procedure	Successful
3	File I/O operation	Successful
4	Database communication	Successful
5	Server/client interaction	Successful
6	Memory usage	Normal
7	System processor usage	Normal
8	Authentication/Authorization	Successful

9. User Manual

9.1 Introduction

The Skill Bites provide a platform for the instructors who is willing to start a paid online course to the students who want to learn specific courses. In these modern days there aren't people who is willing to teach students

9.2 Hardware Requirements

- Intel dual core processor or higher
- 4gb RAM and above
- 40gb hard disk
- Mouse/Keyboard

9.3 Software Requirements

- Language:Nodejs, ReactJS
- User interface design:HTML, CSS, BOOTSTRAP
- Scripting language: javascript, ReactJS
- Operating system:Windows
- Editor: Visual studio code
- Database:MongoDB
- Browsers: google chrome, microsoft edge,etc

10. Conclusion

The project "SkillBites" is a website-based initiative aimed at facilitating instructors in offering paid online courses to students interested in learning specific subjects. With its user-friendly website interface, SkillBites empowers instructors to effortlessly create and deliver high-quality courses while providing them with a seamless means to earn from their expertise. For students, SkillBites offers an immersive learning journey with interactive course materials, video recordings, assignments, and with additional resources. Navigating through the platform is not only straightforward but also enjoyable, fostering an ideal environment for knowledge absorption and skill development. By bridging the gap between instructors and learners, it fosters a vibrant community of knowledge exchange and personal development.

Future Scope:

- In present times where online education is gaining worldwide popularity, SkillBites has major advantage over this trend.
- Providing modules for community discussion tabs enabling instructors to engage in knowledge-sharing with their peers and facilitating student-to-student interactions, SkillBites can foster a vibrant learning community.
- Moreover, Integrating AI tools and real time video communication systems can provide better significance for the growth of SkillBites.